

Production Project Essentials Checklist

A practical checklist of features, files, infrastructure, security, testing, CI/CD, and production details that professional software projects should consider.

1. Website / Frontend Essentials

- Favicon — the small icon shown in the browser tab.
- Page titles (<title>).
- Meta description.
- Open Graph metadata for link previews.
- Twitter/X card metadata.
- robots.txt.
- sitemap.xml.
- Custom 404 page.
- Loading states.
- Error states.
- Empty states.
- Responsive/mobile design.
- Accessibility.
- Dark/light mode when appropriate.
- Proper forms and validation.
- SEO basics.
- Image optimization.
- manifest.json when building a PWA.

2. Security Essentials

- HTTPS.
- Environment variables.
- Never commit secrets or API keys to GitHub.
- Authentication.
- Authorization and permissions.
- Input validation.
- Rate limiting where appropriate.
- Secure cookies/session handling.
- CORS configuration.
- Dependency vulnerability scanning.
- Secure password hashing.
- Protection against common web vulnerabilities.

3. Testing

- Unit tests.

- Integration tests.
- API tests.
- End-to-end (E2E) tests.
- Validation and error-case tests.
- Regression tests.
- Testing of critical user flows.

4. Project / GitHub Essentials

- README.md.
- .gitignore.
- .env.example.
- Clear application tree/folder structure.
- Git branches.
- Pull Requests.
- Code review.
- Branch protection.
- Issue/task tracking.
- Versioning/changelog when appropriate.
- License when appropriate.

5. CI/CD Pipeline

- Git push or Pull Request.
- Install dependencies.
- Lint.
- Format check.
- Type checking.
- Unit tests.
- Integration tests.
- Security/dependency checks.
- Build.
- Deploy to staging.
- Smoke/E2E tests.
- Deploy to production.

6. Production Essentials

- Custom domain.
- SSL/HTTPS.
- DNS configuration.
- Production environment variables.
- Database backups.

- Database migrations.
- Logging.
- Error tracking.
- Monitoring.
- Alerts.
- Health-check endpoint.
- Rollback strategy.
- Deployment strategy.
- Performance monitoring.

7. Database Essentials

- Database schema.
- Migrations.
- Indexes.
- Constraints.
- Backups.
- Restore procedure.
- Connection pooling.
- Proper error handling.
- Do not expose sensitive database information.

8. User Experience Essentials

- Loading feedback.
- Success feedback.
- Clear error messages.
- Retry options where appropriate.
- Handling network failures.
- Handling API failures.
- Empty-data states.
- Invalid form handling.
- Clear behavior while data is loading.
- Undo/confirmation for important actions where appropriate.

9. Analytics & Product Monitoring

- Analytics.
- User activity tracking where appropriate.
- Conversion tracking where appropriate.
- Feature usage.
- Performance metrics.
- Error monitoring.

10. Communication Features

- Welcome email when needed.
- Email verification.
- Password reset.
- Notifications.
- Transactional emails.
- Contact/support system when needed.

11. Maintenance

- Monitor the application.
- Find and fix bugs.
- Apply security updates.
- Update dependencies.
- Maintain the database.
- Optimize performance.
- Collect user feedback.
- Build and release new features.

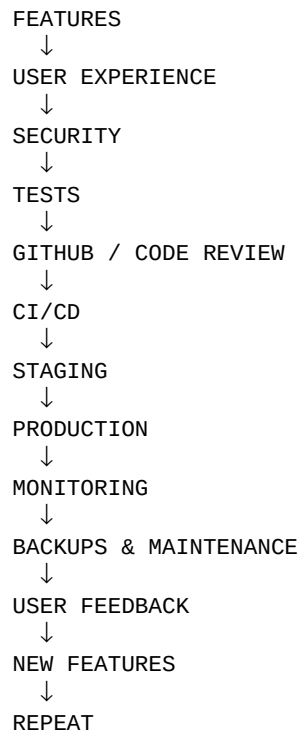
12. Tiny Things Developers Commonly Forget

- favicon.ico
- robots.txt
- sitemap.xml
- .env.example
- README.md
- 404 page
- 500/error page
- Loading states
- Empty states
- Error states
- Open Graph image
- Page metadata
- Health-check endpoint
- Database backups
- Monitoring
- Error tracking
- HTTPS
- CORS configuration
- Rate limiting
- Security headers
- Accessibility

- Mobile responsiveness

Useful Mental Model

A professional application is more than its visible features.



Key idea: the small supporting details—metadata, favicons, error states, security headers, backups, monitoring, documentation, and deployment safeguards—often distinguish a project that merely works from one that is ready to operate reliably in production.